AKTU B.Tech Examination — Answer Key

Subject: Design and Analysis of Algorithms

Answers are included only when available in the source analysis. Missing solutions are flagged for verification.

SECTION A

1. Name the sorting algorithm that is most practically used and also write its Time Complexity

Answer: Answer/solution is not available in the loaded source data. Verify and add a correct solution before using this as an answer key.

2. Differentiate between Backtracking and Branch and Bound Techniques

Answer: Answer/solution is not available in the loaded source data. Verify and add a correct solution before using this as an answer key.

3. Solve the recurrence T (n) = 4T(n/2) + n2 2

Answer: Answer/solution is not available in the loaded source data. Verify and add a correct solution before using this as an answer key.

4. Explain Single source shortest path

Answer: Answer/solution is not available in the loaded source data. Verify and add a correct solution before using this as an answer key.

5. What are approximation algorithms? meant by P (n) approximation algorithms?

Answer: Answer/solution is not available in the loaded source data. Verify and add a correct solution before using this as an answer key.

6. Prove that if n>=1, then for any n-key B-Tree of height h and minimum degree t >~2, h<= log « ((n +1)/2)

Answer: Answer/solution is not available in the loaded source data. Verify and add a correct solution before using this as an answer key.

7. Define principal of optimality. When and how dynamic programming is applicable

Answer: Answer/solution is not available in the loaded source data. Verify and add a correct solution before using this as an answer key.

SECTION B

1. What is Minimum Cost Spanning Tree? Explain Kruskal's Algorithm and Find MST of the Graph. Also write its Time-Complexity

Answer: Answer/solution is not available in the loaded source data. Verify and add a correct solution before using this as an answer key.

2. Explain HEAP-SORT on the array. Illustrate the operation of HEAP-SORT on the array A= {6, 14, 3, 25, 2, 10, 20, 7, 6}

Answer: Answer/solution is not available in the loaded source data. Verify and add a correct solution before using this as an answer key.

3. What is dynamic programming? How, is this approach different from recursion? Explain with example. Fa

Answer: Answer/solution is not available in the loaded source data. Verify and add a correct solution before using this as an answer key.

4. Show the results of inserting the keys F, S, Q, K, C, L, H, T, V, W, M, R, N, P, A, B, X, Y, D, Z, E in order into an empty B-tree. Use t=3, where t is the minimum degree of B- tree. 10

Answer: Answer/solution is not available in the loaded source data. Verify and add a correct solution before using this as an answer key.

5. Solve the following By Recursion Tree Method T(n)=n+T(n/5)+T(4n/5)

Answer: Answer/solution is not available in the loaded source data. Verify and add a correct solution before using this as an answer key.

SECTION C

Q1

(a) Write an algorithm for insertion of key in the Red-Black Tree. Discuss the various cases for insertion of key in red-black tree for given sequence of key in an empty red-black tree- 5, 16, 22, 25, 2, 10, 18, 30, 50, 12, 1. 10

Answer: Answer/solution is not available in the loaded source data. Verify and add a correct solution before using this as an answer key.

(b) Explain and write an algorithm for union of two binomial heaps and also write its time complexity? 10 5

Answer: Answer/solution is not available in the loaded source data. Verify and add a correct solution before using this as an answer key.

Q2

(a) Describe in detail Knuth-Morris-Pratt string ching algorithm. Compute the pref x function 7 for the pattern ababbabbabbababbabb when the alphabet is ~ = {a,b}

Answer: Answer/solution is not available in the loaded source data. Verify and add a correct solution before using this as an answer key.

(b) What is an approximation algorithm?' ne is meant by P (n) approximation algorithms? Discuss approximation algorithm for Travelling Salesman Problem

Answer: Answer/solution is not available in the loaded source data. Verify and add a correct solution before using this as an answer key.

Q3

(a) Define minimum spanning tree (MST). Write Prim's algorithm to generate a MST for any given weighted graph. Generate MST for the following graph using Prim's algorithm. 10

Answer: Answer/solution is not available in the loaded source data. Verify and add a correct solution before using this as an answer key.

(b) Explain Dijkstra's algorithm to solve single source shortest path problem with suitable example. 10 6

Answer: Answer/solution is not available in the loaded source data. Verify and add a correct solution before using this as an answer key.

Q4

(a) Among Merge sort, Insertion sort and quick sort which sorting technique is the best in worst case. Apply the best one among these algorithms to Sort the list E, X, A, M, P, L, E in alphabetic order

Answer: Answer/solution is not available in the loaded source data. Verify and add a correct solution before using this as an answer key.

(b) Solve the recurrence using recursion tree method: T (n)=T (1/2) + T (W/4) + T (1/8) +0 Sub Code: RCSS02 Paper Id: 110502 Roll No: L | | | | | | | 4

Answer: Answer/solution is not available in the loaded source data. Verify and add a correct solution before using this as an answer key.

Q5

(a) What is the application of Fast Fourier Transform (FFT)? Also write the recursive algorithm for FFT

Answer: Answer/solution is not available in the loaded source data. Verify and add a correct solution before using this as an answer key.

(b) Give a linear time algorithin to determine if a text T is a cycle rotation of another string T* For example, 'RAJA' and 'JARA' are cyclic rotations of each other

Answer: Answer/solution is not available in the loaded source data. Verify and add a correct solution before using this as an answer key.